

Metric & Normed Spaces

Pathak – Chapter 1b: Spaces, Operators, Duality and Compactness

Nonlinear Analysis and Fixed Point Theory

2018

Hilbert Spaces

Definition (Hilbert Space)

A **Hilbert space** $\mathcal{H}$ is an inner product space $(\mathcal{H}, \langle \cdot, \cdot \rangle)$ such that the induced norm $\|x\| = \sqrt{\langle x, x \rangle}$ is complete (every Cauchy sequence converges).

Key Properties:

- Complete inner product space with norm $\|x\| = \sqrt{\langle x, x \rangle}$
- Parallelogram law holds: $\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2)$
- Banach space with additional structure

Examples:

- $\mathbb{R}$ and $\mathbb{C}$ with usual inner products
- $\ell^2 = \{x = (x_i) : \sum_{i=1}^{\infty} |x_i|^2 < \infty\}$ with $\langle x, y \rangle = \sum_{i=1}^{\infty} x_i y_i$
- $L^2[a, b]$ with $\langle f, g \rangle = \int_a^b f(t)g(t)dt$
- $C[a, b]$ with an appropriate inner product (normed but not Hilbert in the sup norm)

When is ℓ_p a Hilbert Space?

Example 1.24: For $p \neq 2$, ℓ_p is NOT a Hilbert space.

Counterexample with $p = 3$: Let $x = (1, -1, 1, 0, \dots)$ and $y = (-1, 1, 1, 0, \dots)$. Then:

$$\|x\| = (1 + 1 + 1)^{1/3} = 3^{1/3}$$

$$\|y\| = (1 + 1 + 1)^{1/3} = 3^{1/3}$$

$$\|x + y\| = (2)^{1/3}, \quad \|x - y\| = (2 + 2)^{1/3} = 4^{1/3}$$

Check parallelogram law:

$$\|x + y\|^2 + \|x - y\|^2 = 2^{2/3} + 4^{2/3} \neq 2 \cdot 2 \cdot 3^{2/3}$$

The parallelogram law **fails** for $p \neq 2$, so ℓ_p is not Hilbert.

Only ℓ^2 is a Hilbert space among $\{\ell_p : p \geq 1\}$

Linear Operators and Functionals

Definition (Linear Operator)

Let X, Y be vector spaces. An operator $A : X \rightarrow Y$ is **linear** if:

- ① *Additive*: $A(x + y) = Ax + Ay$ for all $x, y \in X$
- ② *Homogeneous*: $A(\alpha x) = \alpha A(x)$ for all $x \in X, \alpha \in \mathbb{K}$

Equivalently: $A(\alpha x + \beta y) = \alpha A(x) + \beta A(y)$ for all $x, y \in X, \alpha, \beta \in \mathbb{K}$.

Linear Functional: If $Y = \mathbb{K}$, then A is called a **linear functional**.

Note: A linear operator **need not be continuous** in infinite-dimensional spaces!

Key Operators Example:

- $Ax = \sum_{i=1}^n a_i x_i$ on $\mathbb{R}^n$ (linear functional)
- $Ax = (x_2/2, x_3/4, \dots, x_n/n^2, \dots)$ on ℓ_2 (linear operator)
- $[Ax](s) = \int_0^s x(t)y(t)dt$ on $L_2[0, 2\pi]$ (linear functional)

Bounded Linear Operators

Definition (Bounded Linear Operator)

A linear operator $A : X \rightarrow Y$ between normed spaces is **bounded** if there exists $M > 0$ such that:

$$\|Ax\| \leq M\|x\| \quad \text{for all } x \in X$$

Important Terminology:

- *Bounded* does NOT mean image is bounded
- Rather: bounded sets map to bounded sets
- Equivalently: linear operator is bounded iff continuous

Operator Norm:

$$\|A\|_B = \inf\{M : \|Ax\| \leq M\|x\|, \forall x \in X\} = \sup_{\|x\| \leq 1} \|Ax\| = \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|}$$

Theorem (Theorem 1.13)

For normed spaces X, Y : A linear operator $A : X \rightarrow Y$ is bounded if and only if it is continuous.

Example: Unbounded Linear Operator

Example 1.26: Not all linear operators are bounded!

Let c_{00} = space of finitely nonzero sequences with supremum norm $\|\cdot\|_\infty$.

Define $A : c_{00} \rightarrow \mathbb{R}$ by:

$$Ax = \sum_{i=1}^n ix_i \quad \text{for } x = (x_1, \dots, x_n, 0, 0, \dots)$$

Linear? Yes. $A(\alpha x + \beta y) = \alpha A(x) + \beta A(y)$.

Unbounded? Consider $x_n = (1, 1, \dots, 1, 0, 0, \dots)$ (n ones):

$$\|x_n\|_\infty = 1$$

$$Ax_n = 1 + 2 + \dots + n = \frac{n(n+1)}{2} \rightarrow \infty$$

No M exists such that $|Ax_n| \leq M \cdot 1$ for all n

Key Lesson: Even simple looking operators can be unbounded on infinite-dimensional spaces!

The Space $B(X, Y)$ of Bounded Linear Operators

Definition

Let $B(X, Y)$ denote the family of all bounded linear operators from X to Y .

Vector Space Structure:

- $(A_1 + A_2)(x) = A_1(x) + A_2(x)$
- $(\alpha A)(x) = \alpha A(x)$
- Forms a normed space with $\|A\|_B = \sup_{\|x\| \leq 1} \|Ax\|$

Theorem (Theorem 1.14 — Uniform Boundedness Principle)

If X is a Banach space, Y is a normed space, and $\{A_i\}_{i \in I}$ is a family of bounded operators such that $\{A_i x : i \in I\}$ is bounded for each $x \in X$, then $\{A_i\}$ is uniformly bounded: $\sup_i \|A_i\|_B < \infty$.

Consequence: If Y is also Banach, then $B(X, Y)$ is a Banach space.

Convergence of Operators

Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed spaces, and $\{T_n\} \subseteq B(X, Y)$.

Definition (Three Types of Operator Convergence)

- 1 **Uniform convergence:** $T_n \rightarrow T$ uniformly if $\|T_n - T\|_B \rightarrow 0$
- 2 **Strong convergence:** $T_n \rightarrow T$ strongly if $\|T_n x - T x\|_Y \rightarrow 0$ for each $x \in X$
- 3 **Weak convergence:** $T_n \rightarrow T$ weakly if $|f(T_n x) - f(T x)| \rightarrow 0$ for all $x \in X, f \in Y^*$

Theorem (Theorem 1.15)

If $\{A_n\}$ is a sequence of bounded operators with $A_n x \rightarrow A x$ for each $x \in X$ (strong convergence), then $A \in B(X, Y)$ and $\|A\|_B \leq \liminf_{n \rightarrow \infty} \|A_n\|_B$.

Hierarchy of Operator Convergence



Code: Operator Norm in Python

```
import numpy as np
from scipy.linalg import norm

# Define a linear operator (matrix)
A = np.array([[1, 2], [3, 4]])

# Operator norm  $\|A\|_B = \max \|Ax\| / \|x\|$ 
# For matrices, this is the largest singular value
operator_norm = np.linalg.norm(A, ord=2) # spectral norm
print(f"Operator norm: {operator_norm:.4f}")

# Verify:  $\|Ax\| \leq \text{operator\_norm} * \|x\|$ 
x = np.array([1, 1])
x_norm = np.linalg.norm(x)
Ax = A @ x
Ax_norm = np.linalg.norm(Ax)
ratio = Ax_norm / x_norm

print(f" $\|x\| = {x\_norm:.4f}$ ")
print(f" $\|Ax\| = {Ax\_norm:.4f}$ ")
print(f" $\|Ax\| / \|x\| = {ratio:.4f} \leq \{operator\_norm:.4f\}$ ")
```

Banach Algebra

Definition (Algebra over a Field)

A linear space $\mathcal{A}$ over field $\mathbb{K}$ with multiplication $x \cdot y \in \mathcal{A}$ is an **algebra** if:

- 1 $(xy)z = x(yz)$ (associativity)
- 2 $x(y + z) = xy + xz$ (distributivity)
- 3 $(x + y)z = xz + yz$ (distributivity)
- 4 $\alpha(xy) = (\alpha x)y = x(\alpha y)$ (scalar compatibility)

Definition (Normed Algebra / Banach Algebra)

An algebra $\mathcal{A}$ is a **normed algebra** if it is a normed space with:

$$\|xy\| \leq \|x\|\|y\| \quad \text{for all } x, y \in \mathcal{A}$$

A **Banach algebra** is a complete normed algebra.

Examples:

The Hahn-Banach Theorem (Real Case)

One of the most important theorems in functional analysis.

Definition (Sublinear Functional)

A functional $p : X \rightarrow \mathbb{R}$ is **sublinear** if:

- ① $p(x + y) \leq p(x) + p(y)$ for all $x, y \in X$ (subadditivity)
- ② $p(\alpha x) = \alpha p(x)$ for all $x \in X, \alpha \geq 0$ (positive homogeneity)

Theorem (Theorem 1.16 — Hahn-Banach, Real Case)

Let X be a real linear space, Y a subspace, p a sublinear functional on X , and f a linear functional on Y with $f(x) \leq p(x)$ for all $x \in Y$.

Then there exists a linear extension $F : X \rightarrow \mathbb{R}$ such that:

- $F(x) = f(x)$ for all $x \in Y$
- $F(x) \leq p(x)$ for all $x \in X$

Intuition: Can always extend functionals preserving bounds.

Hahn-Banach for Normed Spaces

Theorem (Theorem 1.17 — Hahn-Banach, Normed Case)

Let X be a real normed space, Y a subspace of X , and $f \in Y^$ a bounded linear functional on Y . Then there exists $F \in X^*$ such that:*

- $F(x) = f(x)$ for all $x \in Y$
- $\|F\|_* = \|f\|_*$

Every bounded functional on a subspace extends to the whole space with preservation of norm.

Immediate Corollaries:

Corollary (Corollary 1.2)

For any nonzero $x \in X$, there exists $f \in X^$ with $f(x) = \|x\|$ and $\|f\|_* = 1$.*

Corollary (Corollary 1.3)

For any nonzero $x \in X$, there exists $f \in X^$ with $f(x) = \|x\|$ and $\|f\|_* = 1$.*

Consequence: Enough bounded functionals to separate points!

Geometric Forms: Hyperplanes

Definition (Hyperplane)

A subset H of linear space X is a **hyperplane** if there exists a nonzero linear functional f and scalar α such that:

$$H = \{x \in X : f(x) = \alpha\}$$

Example 1.28: In $\mathbb{R}$, let $f(x) = \frac{3}{7}x$ and $\alpha = 4$.

$$H = \left\{x : \frac{3}{7}x = 4\right\} = \left\{\frac{28}{3}\right\}$$

This is a “hyperplane” (single point) in $\mathbb{R}$.

Theorem (Theorem 1.18 — Separation Theorem)

Let M be a subspace of normed space X , $x_0 \in X$ with $d(x_0, M) = d > 0$. Then there exists a bounded linear functional $j \in X^*$ with unit norm such that:

- $j(x_0) = d$
- $j(x) = 0$ for all $x \in M$

The Dual Space (Conjugate Space)

Definition (Dual Space)

The **dual space** X^* of normed space X is the space $B(X, \mathbb{K})$ of all bounded linear functionals on X .

Norm on X^* :

$$\|f\|_* = \sup_{x \neq 0} \frac{|f(x)|}{\|x\|} = \sup_{\|x\| \leq 1} |f(x)|$$

Key Fact: If X is Banach, then X^* is also Banach.

Examples of Dual Spaces:

- ① $(\mathbb{R}^n)^* = \mathbb{R}^n$ (isometric isomorphism)
- ② $(\ell_p)^* = \ell_q$ where $\frac{1}{p} + \frac{1}{q} = 1$
- ③ $(c_0)^* = \ell_1$ (sequences summable to 0 pair with ℓ_1)
- ④ $(L_p(\Omega))^* = L_q(\Omega)$ where $\frac{1}{p} + \frac{1}{q} = 1$
- ⑤ $(C(\Omega))^* = M(\Omega)$ (space of Radon measures)

Natural Embedding and Reflexivity

Definition (Natural Embedding Mapping)

For normed space X , define $\varphi : X \rightarrow X^{**}$ by:

$$\varphi(x)(f) = f(x) \quad \text{for all } x \in X, f \in X^*$$

This is called the **natural embedding mapping**.

Properties of φ :

- **Linear:** $\varphi(\alpha x + \beta y) = \alpha \varphi(x) + \beta \varphi(y)$
- **Isometric:** $\|\varphi(x)\| = \|x\|$ for all $x \in X$

Definition (Reflexive Space)

A Banach space X is **reflexive** if the natural embedding $\varphi : X \rightarrow X^{**}$ is surjective (i.e., $\varphi(X) = X^{**}$).

Examples:

- Reflexive: ℓ^p for $1 < p < \infty$, Hilbert spaces, finite-dimensional spaces

Key Theorems on Reflexive Spaces

Theorem (Theorem 1.21 — James' Theorem)

If X is a reflexive Banach space, then every $f \in X^$ attains its norm on the unit ball $\mathbb{B}_X$.*

Theorem (Theorem 1.22)

A Banach space X is reflexive if and only if every $f \in X^$ attains its norm on the unit ball $\mathbb{B}_X$.*

Theorem (Theorem 1.23 — Riesz Representation Theorem)

Let $(\mathcal{H}, \langle \cdot, \cdot \rangle)$ be a Hilbert space and $f \in \mathcal{H}^$. Then there exists a unique $y \in \mathcal{H}$ such that:*

$$f(x) = \langle x, y \rangle$$

Moreover, $\|f\|_ = \|y\|$.*

Consequence: All Hilbert spaces are reflexive.

Remark 1.8: In Hilbert spaces, $\mathcal{H}^* = \mathcal{H}$ (in the sense of isometric isomorphism).

Multilinear Mappings

Definition (Multilinear Mapping)

A mapping $A : X_1 \times X_2 \times \cdots \times X_n \rightarrow Y$ is **multilinear** if it is linear in each variable separately.

Definition (Bounded Multilinear Mapping)

A multilinear mapping A is **bounded** if there exists $\mu > 0$ such that:

$$\|A(x_1, x_2, \dots, x_n)\| \leq \mu \|x_1\| \|x_2\| \cdots \|x_n\|$$

Theorem (Theorem 1.20)

There is an isometric isomorphism between $\mathcal{L}(X_1, X_2; Y)$ and $\mathcal{L}(X_1, \mathcal{L}(X_2, Y))$.

Consequence: Multilinear maps reduce to iterates of linear maps:

$$\mathcal{L}(X_1, X_2, X_3; Y) \cong \mathcal{L}(X_1, \mathcal{L}(X_2, \mathcal{L}(X_3, Y)))$$

Weak Topologies

Definition (Weak Topology on X)

The **weak topology** $\sigma(X, X^*)$ on normed space X is the weakest topology making all functionals $f \in X^*$ continuous.

Weak Neighborhoods: A set $G \subseteq X$ is open in weak topology if for each $x \in G$, there exist $f_1, \dots, f_n \in X^*$ and $\epsilon_1, \dots, \epsilon_n > 0$ such that:

$$\{y \in X : |f_i(x) - f_i(y)| < \epsilon_i, i = 1, \dots, n\} \subseteq G$$

Definition (Weak Convergence)

A sequence $\{x_n\}$ **weakly converges** to x (written $x_n \rightharpoonup x$) if:

$$f(x_n) \rightarrow f(x) \quad \text{for all } f \in X^*$$

- Weak topology is weaker than norm topology (fewer open sets)
- Strong convergence implies weak convergence (not conversely!)

Properties of Weakly Convergent Sequences

[Proposition 1.9] The weak limit is unique.

[Proposition 1.10] If $x_n \rightarrow x$ strongly, then $x_n \rightharpoonup x$ weakly.

Example 1.31: Converse fails in ℓ_2 . Let $e_n = (0, \dots, 1, 0, \dots)$ (1 in n -th position).

$$\|e_n\|_2 = 1 \quad (\text{NOT converging to 0 in norm})$$

$$\langle e_n, y \rangle = y_n \rightarrow 0 \quad (\text{converges weakly to 0 for any } y \in \ell_2)$$

$\Rightarrow$ Weak convergence does NOT imply strong convergence!

[Proposition 1.11 — Weak convergence in ℓ_p] For $1 < p < \infty$, $x_n = (x_n^{(i)}) \rightharpoonup x = (x_i)$ in ℓ_p iff:

- ① $\{x_n\}$ is bounded: $\sup_n \|x_n\|_p < \infty$
- ② Coordinatewise convergence: $x_n^{(i)} \rightarrow x_i$ for all i

Weak* Topology and Schur Property

Definition (Weak* Topology on X^*)

The **weak* topology** $\sigma(X^*, X)$ is the weakest topology making all evaluation functionals $e_x(f) = f(x)$ continuous.

Weak* Convergence: $f_n \xrightarrow{w^*} f$ in X^* iff $f_n(x) \rightarrow f(x)$ for all $x \in X$.

Definition (Schur Property)

A normed space X has the **Schur property** if every weakly convergent sequence in X converges in norm.

Lemma (Lemma 1.1 — Schur's Lemma)

ℓ^1 has the Schur property: if $x^n \rightharpoonup 0$ in ℓ_1 , then $\|x^n\|_1 \rightarrow 0$.

Crucial Relationships:

- $(c_0)^* = \ell_1$ and $(\ell_1)^* = \ell_\infty$
- $c^* = \ell_1$ (sequences converging to constant pair with ℓ_1)

Convergence in the Dual Space

Let X be a normed space and $\{f_n\} \subseteq X^*$.

- ① **Norm convergence:** $f_n \rightarrow f$ if $\|f_n - f\|_* \rightarrow 0$
- ② **Weak convergence on X^* :** $f_n \rightharpoonup f$ (weak) if $(f_n - f, g) \rightarrow 0$ for all $g \in X^{**}$
- ③ **Weak* convergence on X^* :** $f_n \xrightarrow{w^*} f$ if $f_n(x) \rightarrow f(x)$ for all $x \in X$

Relationships:

Norm convergence $\Rightarrow$ Weak convergence $\Rightarrow$ Weak* convergence

[Proposition 1.12] Let C be a convex subset of normed space X . Then C is weakly closed iff C is norm closed.

Intuition: For convex sets, weak and norm closure coincide!

Code: Weak Convergence Example

```
import numpy as np

# Weak convergence in  $l^2$ :
#  $x_n = (0, \dots, 1, 0, \dots)$  (1 in  $n$ -th position)
#  $x_n$  weakly converges to 0 but  $\|x_n\|_2 = 1$  (no strong convergence)

def weak_convergence_l2(N=100):
    """Demonstrate weak vs strong convergence in  $l^2$ """

    # Generate sequences
    for n in [10, 50, 100, 200]:
        # Standard basis vector  $e_n$ 
        x_n = np.zeros(n)
        x_n[n-1] = 1.0 # 1 in  $n$ -th position

        # Strong norm
        strong_norm = np.linalg.norm(x_n)

        # Weak convergence test: inner product with arbitrary  $y$ 
        y = np.random.randn(n)
        weak_value = np.dot(x_n, y)

        print(f"n={n:3d}:  $\|x_n\|_2 = \{strong\_norm:.4f\}$ , "
              f" $\langle x_n, y \rangle = \{weak\_value/np.linalg.norm(y):.6f\}$ ")

    print("\n $\|x_n\|$  stays at 1.0 (no strong convergence)")
    print(" $\langle x_n, y \rangle / \|y\| \rightarrow 0$  (weak convergence)")
```

Compactness in Metric Spaces

Definition (Compact Metric Space)

A metric space X is **compact** if every open cover has a finite subcover. Equivalently, every sequence has a convergent subsequence.

Definition (Totally Bounded (Precompact))

A subset C of metric space X is **totally bounded** if for each $\epsilon > 0$, there exists a finite set $\{x_1, \dots, x_n\} \subseteq X$ such that:

$$C \subseteq \bigcup_{i=1}^n B(x_i, \epsilon)$$

[Proposition 1.15] For a metric space X , the following are equivalent:

- 1 X is compact
- 2 Every sequence has a convergent subsequence
- 3 X is totally bounded and complete

Compactness Properties

Theorem (Theorem 1.24)

The continuous image of a compact metric space is compact.

Theorem (Theorem 1.26)

A continuous mapping from a compact metric space to another metric space is uniformly continuous.

[Proposition 1.16] Let C be a subset of a complete metric space. Then C is compact iff C is closed and totally bounded.

[Proposition 1.17] A normed space X is finite-dimensional iff every closed and bounded subset is compact.

Key Observation in Infinite Dimensions:

- Closed unit ball $\mathbb{B}_X = \{x : \|x\| \leq 1\}$ is NOT compact in infinite-dim spaces
- Example: $\mathbb{B}_{\ell^2} = \{x : \|x\|_2 \leq 1\}$ is closed and bounded but NOT compact
- Proof: Standard basis $\{e_n\}$ is in $\mathbb{B}_{\ell^2}$ but has no convergent subsequence $\square$

Weak Compactness

Definition (Weakly Compact / Weakly Sequentially Compact)

- C is **weakly compact** if it is compact in the weak topology
- C is **weakly sequentially compact** if every sequence in C has a weakly convergent subsequence

Theorem (Theorem 1.28 — Eberlein-Šmulian)

In a Banach space, weak compactness and weak sequential compactness are equivalent.

Theorem (Theorem 1.29 — Kakutani's Theorem)

Let X be a Banach space. The closed unit ball $\mathbb{B}_X = \{x : \|x\| \leq 1\}$ is weakly compact iff X is reflexive.

Theorem (Theorem 1.33 — Banach-Alaóglu)

The closed unit ball $\mathbb{B}_{X^}$ in the dual space X^* is weak* compact.*

Mazur's Theorem and Convexity

Theorem (Theorem 1.27 — Mazur's Theorem)

The closed convex hull $\overline{\text{co}}(C)$ of a compact set C in a Banach space is compact.

Key Lemma: For reflexive Banach spaces:

Theorem (Theorem 1.30)

A Banach space X is reflexive iff every closed convex bounded subset is weakly compact.

Theorem (Theorem 1.31)

Let X be a reflexive Banach space and $C \subseteq X$. Then C is weakly compact iff C is bounded (in strong topology).

- ① Strong convergence $\Rightarrow$ weak convergence
- ② Weak convergence $\Rightarrow$ boundedness
- ③ Reflexive $\Rightarrow$ weakly complete (weak Cauchy sequences converge)

Characterization of Reflexivity

Theorem (Theorem 1.32)

For a Banach space X , the following are equivalent:

- ① X is reflexive
- ② $\mathbb{B}_X$ is weakly compact
- ③ Every bounded sequence has a weakly convergent subsequence
- ④ X^* is reflexive
- ⑤ For any $f \in X^*$, there exists $x \in \mathbb{B}_X$ such that $f(x) = \|f\|_*$
- ⑥ Every closed convex bounded subset of X is weakly compact
- ⑦ $\sigma(X^*, X) = \sigma(X^*, X^{**})$ (weak and weak* topologies on X^* coincide)

Summary Table:

Space	Reflexive?	Remarks
ℓ^p ($1 < p < \infty$)	Yes	Hilbert when $p = 2$
ℓ^1, ℓ^∞	No	Dual pairs
c_0	No	Dual is ℓ^1

Summary and Key Takeaways

Chapter 1b Main Topics:

- 1 **Linear Operators:** Definition, boundedness, continuity equivalence
- 2 **Hahn-Banach Theorem:** Extends functionals with norm preservation
- 3 **Dual Spaces:** X^* , operator norms, duality pairings
- 4 **Weak Topologies:** Weaker than norm topology, useful convergence notion
- 5 **Reflexivity:** When $X = X^{**}$, relates to weak compactness
- 6 **Compactness:** Different in finite vs infinite dimensions

Essential Definitions:

- Banach algebra, reflexive space, weak convergence, Schur property
- Natural embedding, dual space, operator norm, sublinear functional

Most Important Theorems:

- Hahn-Banach (existence of extending functionals)
- Banach-Alaoglu (weak* compactness of unit ball)
- Kakutani-Smulian (reflexivity and weak compactness)
- Riesz representation (Hilbert space duality)

Further Reading and Applications

Next Topics (Chapter 1c onwards):

- Optimization with weak topologies
- Fixed point theorems using compactness
- Existence of solutions using Hahn-Banach

Applications of Hahn-Banach:

- Separation of convex sets
- Characterization of elements by functionals
- Extension of linear constraints
- Minimax theorems

Practical Considerations:

- In finite dimensions: all notions collapse (Heine-Borel, reflexivity automatic)
- In infinite dimensions: weak topology essential (fixed points, variational problems)
- Computational: weak convergence harder to verify but weaker assumption

